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RESEARCH

Predicted motion pressure—metricizing pressure created by pass rushers in the NFL and predicting their motions using weighted K-nearest neighbors machine learning models

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Abstract
A successful pass rush has traditionally only been able to be measured by one of these three outcomes: a sack, a hit, or a hurry-up, which has resulted in pressure being a binary variable. In reality, pass rush is an intricate and rapid part of American football, which is why a more precise metric to evaluate pressure is desired, consequently allowing for more in-depth analysis of both players' and teams' performances, as not only the occurrence, but also the amount of pressure created during a play is of interest and can be vital for performance analytics. In this paper, a weighted k-nearest neighbors (wKNN) machine learning model is used to produce such a metric, returning a percentage of pressure created for every pass rusher at any given moment during a play, and is able to predict the binary occurrence of pressure on a play with over 91% accuracy. Additionally, this wKNN is also used to predict the motion of the pass rusher. The pressure created by the predicted motion is then directly compared with the true pressure, allowing for a concrete analysis of a pass rusher's decision-making compared to the league's average.

Keywords Artificial intelligence · American football · Sports analytics

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1 Introduction

1.1 Data in the NFL

With the vast advancements of technology since the turn of the millennium, both the quality and quantity of data within all sports has increased drastically (Assuncao and Pelechrinis 2019). Coaches and players are looking for any possible—no matter how incremental—improvements to gain the edge over their competition. With this in mind, the National Football League (NFL), in cooperation with NFL Next Gen Stats (NGS) and Amazon Web Services (AWS), started implementing a new technology in 2014, for better and more precise tracking and data collection (The National Football League 2023). American football, unlike its European counterpart, is a static sport from a play-by-play perspective, which makes the sport interesting for data scientists. Similar to tennis, each play has a clear and defined start and end, from pre-snap formation, to the ball snap, to one of many end-of-play scenarios (e.g. a touchdown, tackle, incompletion, etc.), allowing for analysts to group not only plays, but also players, as each position has a clear task that is repeated over and over again each play. One important area within the game of American football is the pass rush and the accompanying pressure exerted by the pass rushers. A definition of pressure in American football will be given in Sect. 1.2. By using advanced data science and statistical methods, more intricate findings can be made within play-by-play data regarding the performance of the players and teams. Metricizing these results creates the possibility for the players and coaches to better analyze performances and assist in the improvement of the team's performance. One of these statistical methods is the (weighted) k-nearest neighbors (wKNN) machine learning model, which will be explained in Sect. 2.1. An overview of how wKNN will be used to accommodate the desire for more precise ways of measuring pressure during pass rush will be given in Sect. 1.3.

1.2 Definition of pressure

In American football, for every play, the team in possession of the ball is called the offense, while the opposing team is called the defense. The goal of the offense is to advance the ball along the field towards the end zone of the defense, either by throwing the ball (pass play), or by running with the ball (run play). In this paper, only pass plays were considered and studied. During a pass play, the offense can be separated into the following position groups: the passer (Quarterback, QB), who is in possession of the ball and aims to throw the ball to one of their teammates, the pass receivers (Wide Receivers, WRs, and Tight Ends, TEs), whose goal is to catch the ball and advance it, and the pass block (offensive line), who are the players protecting the QB from the defensive players attempting to get to and tackle the QB. Similarly, the defense can be separated into two position groups: the pass block (Cornerbacks, CBs, and Safeties), who attempt to stop the ball from reaching its intended target (i.e. the pass receiver), or even to intercept the ball, and the pass rush (the defensive line), who, as mentioned earlier, attempt to "rush" towards the QB with the goal to tackle them. An example of the formations of the offense and defense pre-snap for one play

can be seen in Fig. 1. Here, only three defensive players are involved in the pass rush. The two defensive players who lined up across from the offensive line, but did not engage in pass rush, but instead dropped back to aid in the pass defense are referred to as Linebackers, who can be used for both pass rush, as well as pass defense. In this case, they "faked" (or "showed) that they were rushing, which is used as a tactic to confuse the QB and influence their decision making with regard to the throw. Similarly, for pass plays, the player that here lined up to the right of the QB (the Running Back, RB) can be used as an additional pass blocker, or, like in this example, can run a route as a pass receiver. Not every pass play will have a RB on the field. Plays without a RB are referred to as an "empty backfield", which indicates to the defense that the QB will either pass or run with the ball themselves. Keeping the RB on the field for designed pass plays forces the defense to additionally anticipate the possibility of a run play. The act of the pass rush and the position groups involved in it will be the main focus of this paper.

From the perspective of a pass rusher, the ideal/most successful outcome of a pass rush during a pass play is for them to break through the offensive line, reach the QB, and tackle them to the ground before they are able to pass the ball to the intended receiver. Such an event is labeled as a "sack". Should the pass rusher not be able to sack the QB, but still manages to make contact with the QB and therefore interfere with the motion of the pass and as a result disturb it, the pass rusher is said to have achieved a "hit". If, as a result of a pass rusher breaking through the offensive line, the QB has to move from their original position or is forced to hurry up with their throwing motion simply because of the mere presence of the pass rusher, the pass rusher is considered to have achieved a "hurry up" on the play. All of these events, in which the QB is not able to complete the passing motion without disturbance, are considered as pressure having been created by the pass rusher on the play. In this case, pressure has two distinct meanings: on the one hand, the physical pressure created by a defensive player, on the other, the mental pressure and potential repercussions of a successful pass rush experienced by the QB due to severe injuries (Sosnowski and Brosnan 2023). These two forms of pressure go hand in hand, which can

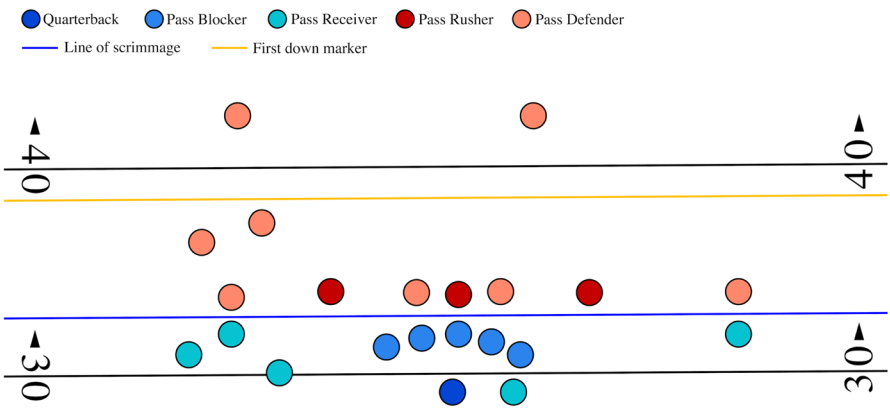

Fig. 1 Birdseye view of a play pre-snap in American football, displaying the different roles players take on during a play

be observed when the QB is forced to hurry up with the pass or has to move to avoid a sack. Naturally, a QB aims to avoid being sacked from a competitive standpoint, as a sack results in a loss of yardage for the offense, making the game more difficult in the subsequent plays. At the same time, the QB is aware of the imminent contact that can be created by the pass rush and the potential consequences that can result from it. While the damages to the body resulting from an injury are one part of the mental pressure on the QB, the financial implications/losses from not being able to compete, as well as the competitive disadvantages that the QB's team suffers from losing one of their most important players, should not be underestimated.

1.3 Need for more precise metrics

While sacks, hits, and hurry ups have already been established to be the results of plays in which pressure occurs, these events are merely categorical for each pass rusher on each play: either a pass rusher has created pressure on the QB, or they have not. Consequently, there is no clear and precise metric that establishes exactly the amount of pressure a pass rusher was able to create at any given moment. As an example, on a play, a pass rusher was able to beat their assigned offensive lineman and had an open lane towards the QB, forcing the QB to roll out of the pocket (the area shaped like a parabola formed by the offensive linemen in which the QB initially stands), and therefore creating a hurry up, but is then stopped by a different offensive lineman, allowing for the QB to gather themselves and complete the pass. In the play-by-play data, the hurry up will then be denoted, but not the exact time when the hurry up actually occurred. Turning pressure into a continuous variable that can be extracted at any given moment of a play, rather than a categorical variable, would allow for far more intricate analysis of plays, as well as of an individual player's and the team's performances, which as a result could have implications regarding strategizing during pre-game preparations or to make in-game adjustments. The existence of a precise measurement of pressure created would allow players to gain a better understanding at which points during a play they performed well, and at which points they could have improved, while at the same time indicating moments during the play that might have created some pressure, but not enough to have been categorized as such in the play-by-play data.

While the NFL, NFL Next Gen Stats and AWS provide a vast amount of data, certain features had to be created, which are in-line with pressure exerted on the QB. With all players being tracked every tenth of a second, all given and created features were available for each player every tenth of a second. In this paper, a data point would represent one player, on one play, at a moment lasting one tenth of a second. Some of the features of these data points relevant to pressure are provided in the data set, such as the speed and orientation of the player at that moment lasting one tenth of second, while other features, such as the x- and y-offsets of the pass rusher to the QB and the pass blockers, had to be created via computation. All of these data points could then be used within a weighted k-nearest neighbors (wKNN) model, with the 13 created features each representing one dimension in the 13-dimensional feature space of the wKNN model. Since it is known for each data point whether or not it is part of a play that created pressure (i.e. a sack, a hit, or a hurry), this information

could be used as a binary categorization. A new data point could then be placed into this space and based on its neighbors, could be categorized to have (not) been part of a play that created pressure. The probability of this point falling into the "pressure-created" category would then represent the likelihood at this moment, that the player would create pressure on the play. This numerical value is proposed as a new metric for pressure evaluation. Additionally, a second metric was created by using the same wKNN model, but rather than predicting the pressure, the motion that the pass rusher would take in the next tenth of a second was predicted. The predicted location, along with speed and orientation, could then be used to generate all pressure-related variables from the predicted position. Putting these predicted variables into the original wKNN would then return a pressure probability for the predicted motion of the pass rusher, representing an entirely separate metric, which has been named Predicted Motion Pressure (PMP). Due to the nature of wKNNs, PMP effectively represents the average motion any player would have taken, as the model compares any given game-state with its k-nearest (most similar) neighbors and computes a weighted average of these data points, which in this case results in a prediction of the motion of a player. This allows for a direct comparison between the actual pressure created by the true motion of a pass rusher at any moment with the hypothetical pressure created from the predicted motion the pass rusher could have taken, immediately giving insights whether the true motion of the pass rusher was better or worse than the average motion (Capellera et al. 2024).

1.4 Related work on pressure and ghosting

1.4.1 Related work on pressure

Over the last few years, more advanced measures have been introduced to move away from classification-based metrics that evaluate defensive pressure. For instance, stats like Time in Pocket, NGS Get-Off (Hermsmeyer 2021), and Pass Rush Win Rate (Burke 2018) improve granularity but remain threshold-based and reduce rich tracking data to simplified outcomes.

Recent work has moved toward continuous, data-driven models. STRAIN, inspired by strain rate in materials science, to measure defensive pressure continuously during a play (Nguyen et al. 2023). STRAIN is defined at each moment by two directly observed features from tracking: the distance between a pass rusher and the QB, and how quickly that distance is shrinking. Intuitively, it quantifies pressure as the closing speed of a defender towards the QB, scaled by proximity. This yields a time-varying pressure curve for every rush, allowing evaluation even on plays without a sack or hit. STRAIN is further aggregated via a multilevel model to estimate each rusher's contribution to pressure on a per-play basis, controlling for situation and team effects. This research is accounted for in NFL's own NGS Pressure Probability metric. This model quantifies, on a per-second basis, the evolving likelihood of pressure using sensor data, rusher-blocker matchups, and QB reaction features-enhancing insights into pass rush effectiveness (NFL Next Gen Stats 2023).

Similarly, a unified framework for automating pass rush statistics has been proposed (Hong et al. 2024). This system first uses machine learning to identify blockers

and rushers and pair them in matchups on each play, since who is actually rushing or blocking is not explicitly given in raw tracking data. Both a rule-based ensemble approach and a graph neural network are explored to solve this identification and matchup problem, achieving high accuracy in determining which players are pass rushers or blockers in a play. Given those pairings, Hong et al. then model pressure probability as a continuous function over the course of a play, which is essentially the likelihood that a pressure outcome occurs at each moment as the play progresses. A tree-based model with engineered features from the tracking data is trained to predict each rusher's pressure probability in fine-grained time steps. The result is a time-series of pressure probabilities and derived statistics that indicate how fast, how long, and how effectively a defender generates pressure during the play.

Lastly, Instantaneous Disruption Probability Increase (IDPI) measures how much a given defensive lineman increases his team's probability of disrupting the pass on a play, compared to an average player in the same situation (Bachelder 2023). This approach uses an LSTM recurrent neural network to estimate, at each moment, the probability of a pass disruption (sack or incompletion) with and without that particular rusher's presence. The difference (the increase in disruption probability) is attributed to the rusher, capturing his impact even if he does not record a traditional stat.

1.4.2 Related work on ghosting

The other important line of relevant research in this context is "ghosting", which refers to simulating how a player should have behaved in a given moment, enabling comparisons between actual and optimal performance. The idea originated in basketball (Lowe 2013), and was formalized with the introduction of data-driven ghosting via deep imitation learning (Le et al. 2017). Their method trained models on tracking data to generate league-average defensive trajectories in soccer, allowing analysts to contrast actual defender movement with simulated "ghost" defenders and measure the impact on expected goals.

The technique has since evolved across sports. In basketball, interactive tools have been created to sketch offensive movement and generate ghost defenders in response (Seidl et al. 2018). In American football, average defender trajectories are simulated and compared to actual plays to estimate defensive quality, especially in pass coverage scenarios (Schmid et al. 2021).

Most relevantly, the NFL Ghosts framework applies conditional density estimation to generate 2D distributions of expected defender positions at the moment of a catch (Yurko et al. 2025). This method compares real defenders to baseline "ghost" players and quantifies the impact of positioning on yards after catch and expected points. Unlike earlier ghosting models, NFL Ghosts returns full spatial distributions with uncertainty estimates, enabling richer evaluations of defensive behavior.

2 Methodology

2.1 General concepts of (weighted) K-nearest neighbors

The k-nearest neighbors (KNN) algorithm is a supervised, non-parametric learning classifier (Valkenborg et al. 2023). It can be used for both classification and for regression problems, both of which will be present in this paper. With KNN, maximum condition probability is used as a decision rule to predict the classification of a new observation (Kelen and Benczúr 2023). Simply put, for KNN, the training data is placed in an n-dimensional space, with n representing the amount of features of the data. A new observation is then put in this space and via proximity to the k-nearest neighbors, the classification of the new observation is then determined. Generally, KNN is based on "majority voting", meaning more than 50% of the k-nearest neighbors need to be of a classification for the new observation to be classified as the same (Bzdok et al. 2018). This threshold can be chosen freely, though, as one way of parameter tuning (Zhang et al. 2017). An example of a simple KNN with two features and two classes can be seen in Fig. 2. Here, three of the five nearest neighbors are of class A, therefore, the KNN also classifies the new observation into class A.

The value for k naturally plays a vital role in the classification process, while other parameters can also be tuned, such as giving weight to the distance to the neighbors (Uddin et al. 2022). In this case, neighbors that are closer in proximity have a higher influence on the prediction compared to neighbors with greater distance. As an example, if k=19, and 9 data points of classification A are relatively close to the new observation, while 10 data points of classification B are relatively far from the new observation, by default, the KNN model would classify the new observation as B, because more than half of the k nearest neighbors are class B, while with a weighted distance, it would be classified as A. Weighted KNN can generally improve the classification accuracy, especially with regard to the effects of noisy data (Halder et al. 2024). A prediction $\hat{y}$ for a query point x_q using a weighted KNN with two classes c_A and c_B can be expressed as:

$$\hat{y} = \begin{cases} c_A & \text{if} \quad P(c_A|x_q) > P(c_B|x_q), \\ c_B & \text{otherwise.} \end{cases} \tag{1}$$

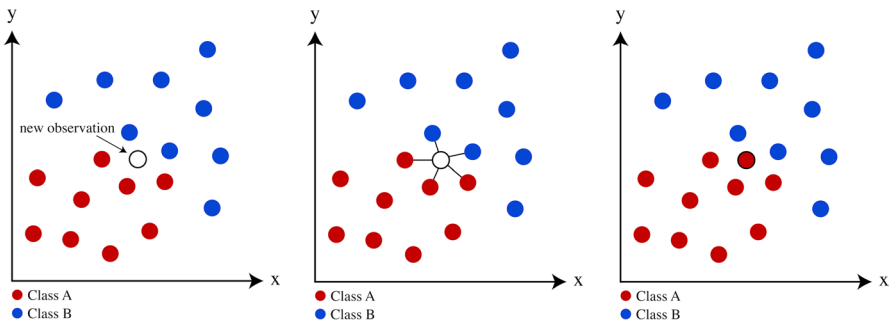

Fig. 2 Visualization of a KNN with two classes and k=5

where $P(c_A|x_q)$ and $P(c_B|xq_)$ represent the probability that the query point x_q is of class c_A and c_B respectively. These probabilities can be expressed as:

$$P(c_A|x_q) = \frac{\sum_{i=1}^{k} K(d_i) \cdot I[y_i = c_A]}{\sum_{i=1}^{k} K(d_i)} \tag{2}$$

and

$$P(c_B|x_q) = \frac{\sum_{i=1}^{k} K(d_i) \cdot I[y_i = c_B]}{\sum_{i=1}^{k} K(d_i)} \tag{3}$$

where *k* is the number of nearest neighbors, y_i is the target value of the *i*-th nearest neighbor, $I[y_i = c_A]$ and $I[y_i = c_B]$ are the indicator functions that equal 1 if the target y_i of the *i*-th neighbor belongs to the class c_A or c_B respectively, and equals 0 otherwise, and $K(d_i)$, which is the kernel of the inverse-distance function, and can be described as:

$$K(d_i) = \frac{1}{d_i + \epsilon} \tag{4}$$

where $d_i = \|x_q - x_i\|$ represents the distance between the query point x_q and the *i*-th neighbor x_i, and ϵ is a small positive constant added to avoid division by zero.

One major advantage of KNN that was of importance in this paper is the fact that the probability of each prediction would be extrapolated straightforwardly. In the aforementioned example, with three out of the five nearest neighbors being of class A, the probability that the new observation would be of class A would be equal to 60%, while the probability of the new observation being of class B would be equal to 40%. For regression problems, the same concept applies, but instead of a classification, the average value of the response variable of the k-nearest neighbors is calculated and used to the determine the value of the new observation. An important benefit of KNN models is that the probability of the prediction can easily be extrapolated, which was a key factor in the creation of the pressure-related metrics in this paper, which will be elaborated on in Sect. 3.6 (Gweon et al. 2019).

2.2 Data

The data used for this paper was provided by the NFL, NFL Next Gen Stats, and Pro Football Focus (PFF) via the NFL Big Data Bowl 2023, which was hosted on Kaggle, and is publicly available (Kaggle 2022). The data contains information for the first eight weeks of the 2021 NFL season, tracking every outcome of every play, as well as precise telemetry for every player involved on the play. There are 12 CSV files, which can be divided in five different categories: game, play, player, PFF Scouting, and tracking data. With the focus laying on pressure on pass plays, only dropback pass plays were included in the data.

The game data contains all games played in the first eight weeks of the 2021 NFL season with the names of the teams that have played in each game. Each game is given a unique *gameId*, by which games can be identified through the different data sets.

The play data contains play-level information for each play of each game, which includes a description of the play. As an example: (13:33) (Shotgun) T. Brady pass incomplete deep right to C. Godwin. This description contains the time on the game clock, the formation of the offense and (in the event of a pass play), the name of the passer, the intended receiver and where the ball was thrown to. Each play can be identified by a *playId*, as well as the corresponding *gameId*, for which the *playId* will be unique for, meaning that identical *playId* may appear for different games.

The player data contains player-level information for each player that has been tracked at least one time throughout the games. Each player can be identified with a unique *nflId*.

For each game and play, player-level scouting information is provided by PFF for all 22 players on the field. This includes information such as which role a certain player has (e.g. pass rush), whether a player created or allowed a sack/hit/hurry on the QB and, if applicable, which player they faced in a matchup. Each player can be identified via the unique *nflId*, the *playId* and the *gameId*.

For each player and the football, in each game and play, specific tracking data was recorded every tenth of a second, which is considered a frame. This tracking data includes the player's and the football's x-coordinates, y-coordinates, speed, acceleration, and orientation for every frame of the play. Each player can be uniquely identified via their *nflId*, and corresponding *playId* and *gameId*.

Teams in the NFL have their bye-week (the one week during the season in which the team does not compete) between week 6 and week 14, meaning that for weeks 1–5, with 32 teams in the NFL, there were 16 games per week, 14 in week 6, 13 in week 7 and 15 in week 8. Naturally, this is reflected in the the amount of pass plays that were ran by the offense, as over the first five weeks, the average of dropback pass plays per game was at 1120.8, which dropped to 1004 in week 6917 in week 7 and 1032 in week 8. This comes out to approximately 70.14 passing plays per game, or 35.07 per team, on average. A visualization of this distribution can be seen in Fig. 14 in Appendix A. The Kansas City Chiefs were leading this statistic with 42.375 dropback passing plays on average over the first 8 weeks, while the Green Bay Packers were trailing with 28.25 dropback passing plays on average. As mentioned earlier, the entire focus here is on dropback passes, which are only one type of pass, albeit the most common at that (Wilkinson 1993). The difference in amounts of dropback passes amongst the teams can be explained by the type of offense that a team runs, which often correlates to the playing style of the team's QB. A team with a mobile QB with a strong arm (like Kansas City's Patrick Mahomes) is prone to build their offense around him, profiting from efficient pass play, while teams with higher focus on run plays, or preferring to have the QB in the Shotgun formation (in which the QB does not have to drop back), would naturally score lower on this list. Additionally, a decisive factor within the play-calling, is the strength of the offensive line against the pass rush. An offensive line being able to withstand pressure from the pass rush well, is naturally more inclined to continue with passing the ball. The longer an offensive

line is able to withstand the pressure, the more time the QB has to "read" through his receiving options and find the one with the best chance to be able to complete the pass (Michelman and Walter 2019).

In total, 8557 dropback pass plays took place during the first eight weeks of the 2021 NFL season, 3177 (37.13%) of which had a pressure event occur during it. 543 of these plays resulted in the sacking of the QB, equating to 6.35% of all plays, and 17.09% of plays that contained pressure. On these 8557 recorded plays, a total of 36,362 individual pass rushes had taken place. 4232 (11.64%) of these pass rushes resulted in pressure. 2824 (7.77%) of the pass rushes resulted in a hurry up, 823 (2.26%) resulted in a QB hit, and 585 (1.61%) resulted in a sack. There is no overlap between those events, a pass rusher had exclusively created at most one of these events during a pass rush. The distribution of pressure occurrences for plays as well as for individual pass rushes can be seen in Fig. 3. Among all 8557 recorded plays, the football left the QB's hands in the form of a pass 7565 times. 4620 (61.07%) of these passes were completed to the WR, 2755 (36.42%) were deemed incomplete, meaning that the pass was caught by neither a Receiver, nor by the opposing team, and 190 (2.51%) were intercepted, meaning the opposing team caught the ball, giving possession to the defense. These three pass outcomes are mutually exclusive from one another. When looking at these three outcomes in relation to the occurrence of pressure on a play, 3504 of the 4620 completed passes (75.84%) experienced no pressure on the play, while 1551 of the 2755 incomplete passes (56.30%), and 116 of the 190 interceptions (61.05%) occurred without pressure on the play. The distribution of interceptions stands out, as it would be expected that pressure would naturally lead to forcing the QB into more errors. A possible explanation for this may be that on plays in which the QB experiences little to no pressure, they would be more likely to throw a "higher risk/higher" reward pass downfield, which consequently also has a higher probability of being picked off.

The remaining recorded plays consisted of the aforementioned 543 sacks, and 449 QB runs, where the QB keeps possession of the ball and is either tackled for a positive yardage gain, steps out-of-bounds, or scored a touchdown. A Venn diagram displaying the three passing outcome and QB run distributions with regard to pressure on the play can be seen in Fig. 4.

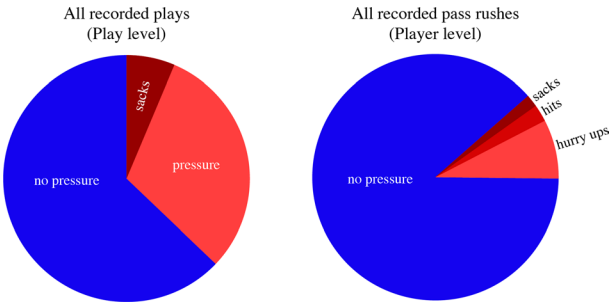

Fig. 3 Distribution of pressure among all recorded plays and all recorded pass rushes

Figure 15 in Appendix A shows the average percentage of plays that had at least one player exerting pressure per game per week. This percentage remains constant at just below 12.5% of plays for seven out of the eight weeks.

2.3 Feature engineering and selection with regard to pressure

It has already been established that a sack, hit, or hurry achieved by a pass rusher on a play equals a high amount of pressure having been exerted by said pass rusher. Naturally, if a pass rusher has beaten their assigned offensive lineman, and has a clear path towards the QB with the potential to make contact, or at least force the QB to move away, high pressure is being exerted on the QB. Similarly, if the pass rusher is being held up by one (or more) offensive lineman at a distance to the QB, little to no pressure is being exerted from this specific pass rusher. Consequently, from the perspective of the pass rusher, this dynamic constellation and all related variables between themselves, the offensive lineman/linemen blocking them, and the QB, is all-decisive in regards to the pressure that is being exerted at any given time.

Here it has to be mentioned that, for any given play, a pass rusher may have one, multiple, or no assigned pass blockers. In this paper, it was decided to only focus on the closest offensive lineman to the pass rusher for the following wKNNs to evaluate pressure. The reason for this being that, if it had been chosen to take several pass blockers into consideration, for all plays in which the pass rusher had less blockers than would have been taken into consideration, the additional pass blockers would have provided no significant data with regard to the evaluation of pressure, and with no/one pass blocker being a substantial percentage of the plays, this additional information increased the risk of overfitting the models. Additionally, when only using the closest pass blocker for the wKNN, the prediction of likelihood of pressure occurring

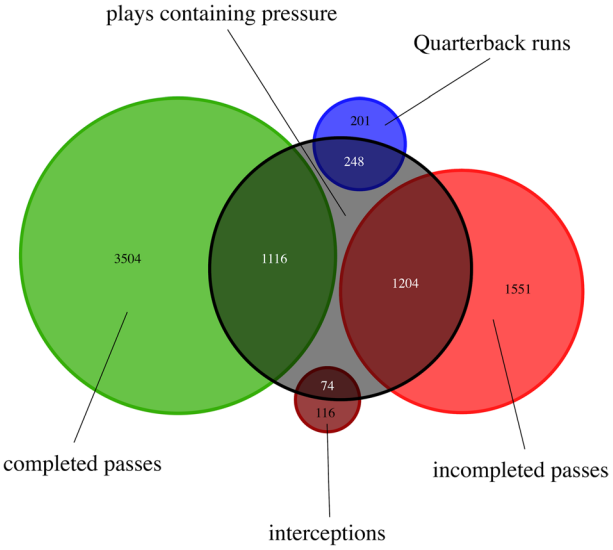

Fig. 4 Venn diagram displaying the distribution of completed and uncompleted passes, as well as interceptions for plays, on which pressure occurred

is not being influenced in the event that multiple offensive linemen are blocking the pass rusher. For example, if a pass rusher is being successfully double-teamed and is therefore not able to get through to the QB, the pressure is naturally low and would be predicted as such, regardless of how many blockers are being considered, since the given situation would correlate with scenarios in which low pressure was generated. Even if the pass rusher would beat one of the two offensive lineman, the second pass blocker would immediately be recognized as the new closest blocker, continuing to keep the pressure low. Only once the pass rusher has beaten both and is able to run towards the QB, the pressure would increase. It can be seen that this dynamic constellation between the QB, the pass blocker, and the pass rusher yield several factors that are important with regard to pressure. Firstly, as previously described, the x-offset, y-offset, and distance between the pass rusher and the pass blocker influence the ability of the pass rusher to get to the QB and therefore also the amount of pressure being created. Similarly, the x-offset, y-offset, and distance between the QB and the pass rusher can be assumed to be crucial with regard to pressure, as well as the angle between the passer, pass blocker, and pass rusher, with the QB at the vertex.

Additionally, the individual speed, and orientation of all three players involved can impact the amount pressure that can be created. Finally, the x-offset and y-offset of the pass rusher compared to their last frame were considered. To summarize, the following 13 features were derived to be used for all wKNN models used in this paper:

- speed of the pass rusher
- orientation of the pass rusher
- x-offset between the pass rusher and the QB
- y-offset between the pass rusher and the QB
- speed of the QB
- orientation of the QB
- x-offset to the closest pass blocker
- y-offset to the closest pass blocker
- speed of the closest pass blocker
- orientation of the closest pass blocker
- angle created between the pass rusher, pass blocker, and QB, with the QB at the vertex
- x-offset of the pass rusher compared to their last frame (directional velocity component in the x-direction)
- y-offset of the pass rusher compared to their last frame (directional velocity component in the y-direction)

For any frame, the speed and orientation of each player are given, as well as the exact x- and y-coordinates. These coordinates were not used in the feature set for the wKNNs, since pass rush can happen on any point of the field. They were instead used to compute the offsets between individual players and to the last frame of the pass rusher. With the first play of every frame not having a previous frame, these offsets of the pass rusher were set to zero. The distances to all pass blockers on the play were computed by using Pythagoras' Theorem, with the x- and y-offsets being known.

With the distances to all pass blockers on a play being known for a pass rusher, the closest one would be kept with all of their features. Additionally, the closest pass blocker could be used to compute the angle between the pass rusher, the closest pass blocker, and the QB, with the latter at the vertex.

For the wKNN model predicting the motion of the pass rusher one tenth into the future, the same features could be used, yet additional computation was necessary for the outcome variables (x-offset, y-offset, and orientation in the next frame). While the offsets had already been computed in the previous step, the offsets in the next frame were of interest. For those variables to become the outcome variables, the relevant offset and orientation variables had to be moved up one row in the data set, so that the wKNN model could identify those values as the one to be predicted for the next frame. This operation would lead to the last frame being matched with a value from an entirely different play, but due to the fact that the last frame would not be needed to create a new frame (since the play is over at that point), all those frames could be deleted from the data set. With the wKNN model then predicting the offsets and orientation of the pass rusher in the next frame, the pass rusher's new x- and y-coordinates could be computed. The coordinates of all other players on the next frame are known, allowing for the creation of a hypothetical frame. With the predicted x- and y-offset values, the predicted traveled distance of the pass rusher compared to the previous frame could again be computed using Pythagoras' Theorem. With speed equaling the change of distance over a certain amount of time, the predicted speed was obtained by dividing the calculated distance by 0.1s (the frequency in which frames were measured). The computation of the necessary features to determine the pressure probability created by the pass rusher in this predicted position is identical with the earlier computation for the true pressure probability.

Creating a high-dimensional space was deemed necessary to account for both the complexity in data, as well as that of pressure exertion. A visualization of these telemetries can be seen in Fig. 5. Additionally, one example frame of a pass rush, with the interactions between the pass rushers and pass blockers can be seen in Fig. 6. The relationships between the individual players can naturally change over the course of the play, as a different pass blocker may become the closest blocker to a pass rusher at a different frame. Each one of the five pass rushers, along with all of their previously mentioned features represents one data point, meaning that for this play, each frame

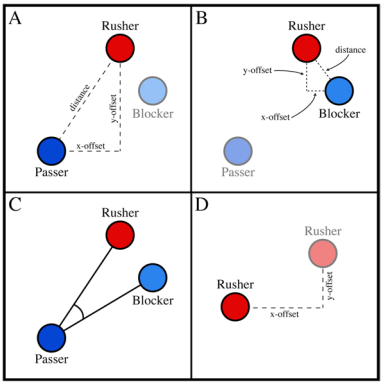

Fig. 5 **A**: x-offset, y-offset and distance between pass rusher and QB **B**: x-offset, y-offset, and distance between rusher and closest Offensive Lineman **C**: angle created between pass rusher, closest Offensive Lineman and QB **D**: x-offset and y-offset of the pass rusher compared to the last frame

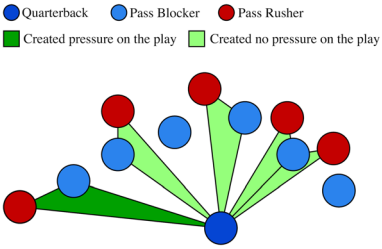

Fig. 6 Visual representation of the interactions between pass rushers and pass blockers in relation to the QB on one still-frame during a pass rush

resulted in five different data points. All 36,362 individual pass rushes therefore output a total of just over 1.5 million data points, with the 13 aforementioned features being present for each data point. For each data point, it is known whether or not the corresponding player created pressure on the play for which that frame is a part of, which is considered the outcome variable for each data point.

3 Results

3.1 Predicting pressure of pass rushers with wKNN

As mentioned in Sect. 2.2, for each pass rusher on each play, it is known whether or not this player was able to create pressure (a sack, hit, or hurry). Therefore, it is known whether each frame (with all relevant and established features) for a pass rusher was part of a play that created pressure. This categorization for each data point was considered to be the outcome variable. With 13 features per data point selected, the feature space for this wKNN would be 13-dimensional. Seven of the eight available weeks were used for training, with the remaining eighth week having been used for testing. Ergo, to retrieve the predictions for each frame of each pass rusher for ever play, the wKNN model would be trained and tested eight separate times. This not only allows for the computation of the likelihood of pressure of every pass rusher at every available frame, which leads to a more in-depth analysis of players, position groups, and teams, but also acts as cross-validation for the model itself. The accuracy of all eight individual models was consistent. A more detailed description of the model validation can be seen in Sect. 3.2. Each data point in the training set, based on the features' values, would then be placed in this 13-dimensional feature space, with the aforementioned categorization of either having been part of a pressure play, or not. A new data point from the testing data set could then be placed into this complete feature space, as all of its features would be identical to the one from the training data points. After using grid search, the highest accuracy was achieved at k=500 neighbors. Since eight different models were trained for each week, this number varied slightly for each model. To allow for uniformity among the models, the same number of k-nearest neighbors was chosen. For each new data point, the wKNN model would compare the frame with its 500 nearest neighbors. The distance of each neighbor to the data point was given weight. Without the added weight, the probability of each prediction would always be a fraction of 500. Typically, for a classification wKNN, the prediction of a new data point is simply its classification based on the majority

of the data point's nearest neighbors, meaning that for a new frame, if the majority of its neighbors would be frames that were part of a play that created pressure, the wKNN would predict this frame to also have been part of a play that created pressure, and vice versa. However, this prediction is merely also a classification and does not provide a precise number to determine the amount of pressure that was created. To achieve this, rather than retrieving the classification, the wKNN would return the likelihood of a new frame belonging to a play containing pressure, based on its nearest neighbors and its distance to those neighbors, i.e. if a frame's majority of closest 500 neighbors would all be frames of plays that created no pressure, the probability for the new frame to be part of a play containing pressure would be relatively low. With this, from this point on the definition of pressure was able to shift from a mere binary metric, to a precise number between 0 and 1, as every frame from every pass rusher on every play could be input into this wKNN and be linked to a percentage signifying the likelihood of the pass rusher—in that very moment—being able to create pressure on the play. This numerical value is interpreted as a proxy for the amount of pressure being created by the pass rusher.

This new metric can be demonstrated in the following example: Fig. 7 shows the pressure created by Indianapolis Colts' DeForest Buckner (nflId 43296) on play 4334 against the Tennessee Titans (gameId 2021103106) in week 8. Buckner's pressure pre-snap (meaning before the ball was passed from the Center to the QB, signifying the moment when the defense is allowed to cross the line of scrimmage), while low compared to the pressure among all frames, is relatively high given that the ball has not been snapped yet, due to the way he lines up on the play, as this constellation is considered relatively likely to have ended in a pressure event by the model, and therefore returning a higher percentage number, ergo more pressure. This means that,

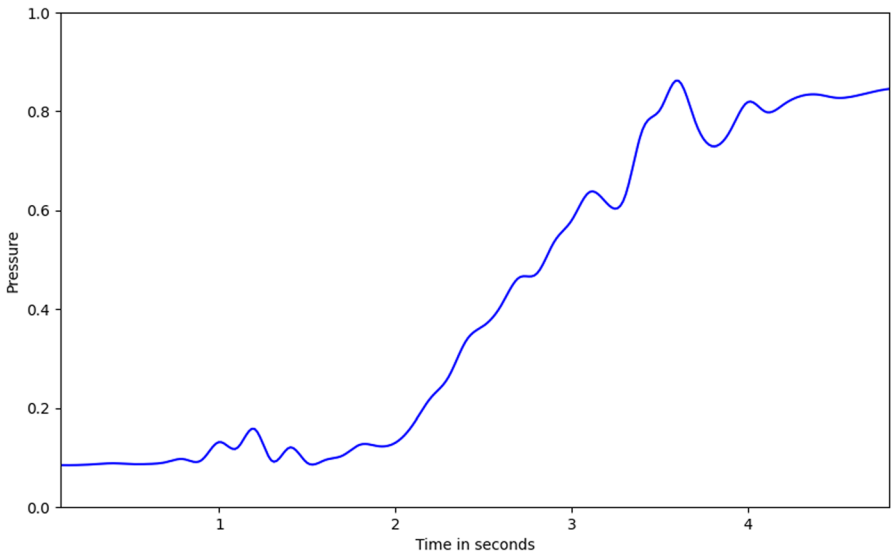

Fig. 7 Pressure created by Deforest Buckner on play 4334 against the Tennessee Titans, resulting in a sack

based on the data points that the wKNN model has been trained on, the 500 nearest neighbors for these data points include an above average amount of data points which were part of pass rushers that induced pressure, based on the way these pass rushers had lined up. The ball is snapped at frame 6 (0.6 s after the tracking started), and while his pressure increases initially, it drops back down and remains level at around 10%. This is due to the fact that Buckner was faced head-on by a pass blocker, whom he was able to beat around frame 20 (1.4 s after the ball snap), giving him a clear path towards the QB of the Tennessee Titans, Ryan Tannehill, gradually increasing his pressure. Buckner's pressure peaked at 86.27%, achieving a sack, and resulting in an 8 yard loss on the play for the Titans, as well as unfortunately injuring Ryan Tannehill in the process.

3.2 Model validation

With 1.5 million total data points available, the quantity used for training of the wKNN models can be considered substantial. Typically, when analyzing the validity of a wKNN model, its prediction accuracy is used, comparing how many predictions were classified correctly compared to the total amount of predictions made. In this case, the classification was not of interest though, leading to the accuracy of the model not representing its true performance. This can be seen in the following scenario: pre-snap, pressure naturally will have to be low, as the pass rusher is standing still, facing the offensive line head-on, with the QB not being in immediate striking distance. The amount of pressure in this case can vary based on the formation of both the offense and the defense, but regardless the overall pressure in this instance will remain low. The prediction probability of the wKNN confirms this, as most of the nearest neighbors of such a frame would also have been part of a play not resulting in a pressure outcome. Logically, the frame's classification would be that the frame was not part of a play resulting in pressure, which may not necessarily have been the case. Later on in the play, the player may have been able to achieve a sack, hit, or hurry, which is not of importance to the pressure percentage on that frame, yet the classification is wrong, hurting the model's performance for this metric. As a result, a different metric needed to be used to find the model's true performance. Instead of validating each data point independently, entire plays were analyzed. For each play, it was known whether or not a pass rusher had created pressure in the form of a sack, a hit, or a hurry up. For the predictions, if one or more frames for a player yielded a likelihood of more than 50% pressure, the player was considered to have created pressure on the play. The default threshold for a wKNN model is typically set to 50%, which is why it was chosen here, but can be considered entirely arbitrary. Unlike when tuning the hyperparameters of a wKNN model, where the changing of the threshold is done to improve the accuracy of the model, the threshold for this evaluation had to be chosen arbitrarily, as the pressure metric signifies an entirely new metric and therefore had nothing concrete to be evaluated against. With this in mind, the model classifies 91.58% of pass rushers correctly on whether or not they had created pressure on a play. The confusion matrix for the individual pass rushes can be seen in Fig. 8. This accuracy score could be increased slightly by increasing the threshold, but considering that the general interest of this research paper was not

to determine whether or not a player had created pressure on a play (as this metric is already available in the form of sacks/hits/hurry ups), but rather to find the precise amount of pressure exerted at any point during a pass rush, this fine-tuning can be considered redundant. Due to the fact that a unique form of measurement of the accuracy of the model had to be created, as there was no metric to test the predictions against, a uniform threshold across all predictions was of importance to get a general sense of the performance of the model.

An additional way of analyzing the validity is to look only at the players that were able to achieve a sack on the play, with the sack being the perfect way of determining pressure. A sack ends the play and finds the pass rusher to be right at the QB. This means that, in a valid model, the pressure on these plays would naturally be very high and the model should predict those plays to have contained pressure. For sack plays only, the model has an accuracy of 92.14%.

Similarly, the strength of the model in the opposite direction can be evaluated. For 92.99% of plays on which the QB was able to complete the pass to one of his receivers, no sack, hit, or hurry up occurred. When evaluating pass rushers on all plays that resulted in a completed pass, regardless of pressure having occurred, the model returned an accuracy of 94.48%. When focusing on the pass rushers on completed pass plays where no pressure occurred, the model yielded an accuracy of 92.50%.

A slightly less scientific, but in this case strong way of validating the model, is the eye-test. Alongside actual video footage of any play, the predicted pressure can be compared to what is occurring on the field. It is fairly easy to give an educated estimate on how much pressure a pass rusher is able to create for the duration of a play, as in whether or not the QB is disturbed by their presence, or the pass rusher is clearly being held up, or even blocked. These observations can then be compared to the numbers provided by the model, which appear to give a realistic representation.

Lastly, it should be mentioned that a model with additional features for the two closest/three closest blockers sees no improvements in accuracy over all plays. There is a difference however on a play-by-play basis. Naturally, when a pass rusher has two blockers, a model considering the two closest blockers is basing its prediction off of more "correct" data. But when a pass rusher only has one blocker, a model considering the two closest blocker will be working with unnecessary, "false" data. The model with only the closest pass blocker will, due to its nature, perform well for

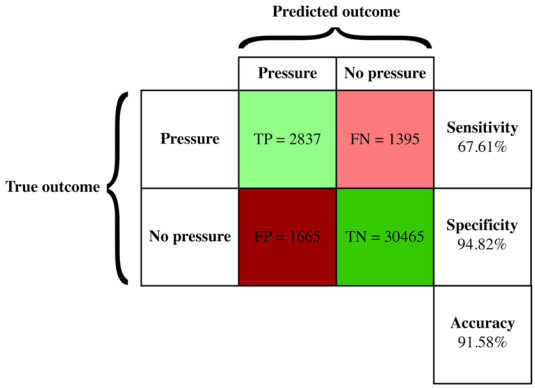

Fig. 8 Confusion matrix for all individual pass rushes, with an accuracy of 91.58%

every case. It could be argued that four separate models could be created for each case that the pass rusher could face (including being unblocked), to achieve the highest amount of accuracy, but the lack of unblocked pass rushes and pass rushes with three blockers would mean insufficient data for some of these models. Therefore, a single model unifying and streamlining the process is believed to have the best overall performance.

3.3 Uncertainty quantification and model calibration

To quantify the uncertainty in the KNN model's predictions, the variability in the predicted likelihood of pressure was estimated using non-parametric bootstrapping (Bates et al. 2015; Yurko et al. 2018). Specifically, two-thirds of the training set were resampled with replacement 1000 times. For each resampled model, predictions were made for frame 36 of Deforest Buckner's pass rush mentioned in Sect. 3.1, which corresponds to the moment of peak pressure during the play. The resulting distribution of 1000 predicted probabilities is shown in Fig. 9. The full model predicted a pressure likelihood of 86.27%, while bootstrapping yielded a standard deviation of 0.0151 and a variance of 0.0002. This relatively low variance suggests that the model's predictions for this frame are stable under perturbations to the training data. Only one data point was used to create an indicator for low variance in the model. Due to the heavy computational load, it would be impractical to run the bootstrapping over the entire dataset.

To assess the model's calibration, prediction binning with 5% intervals was performed. In this procedure, the frequency of positive labels (in this case, plays where pressure occurred) within each bin is evaluated. Each frame's predicted likelihood of pressure for every pass rusher is compared against the true pressure outcome of the corresponding play. For example, in an ideally calibrated model, if, on any given frame, a player is assigned a 10% likelihood of creating pressure on that play, then that player would actually create pressure on approximately 10% of such plays. This relationship can be examined for each probability bin. A visualization of this assessment is shown in Fig. 10. Ideally, the calibration curve aligns closely with the dotted "perfect calibration" line. In this case, the model deviates only slightly in the mid-range probabilities, where it appears to be mildly under-confident, while it remains nearly perfectly calibrated for both the low- and high-probability ranges. With an expected calibration error (ECE) of 1.2% and a maximum calibration error (MCE) of 15.4%, the model can be considered to be overall well calibrated.

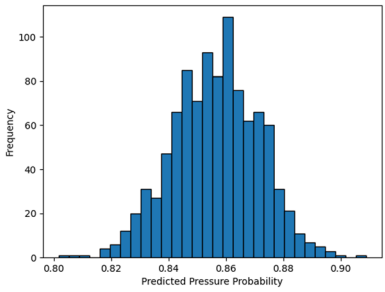

Fig. 9 Histogram of predicted pressure likelihoods for frame 36 of Deforest Buckner's pass rush, obtained from 1000 bootstrap resamples of the training set using two-thirds sampling with replacement

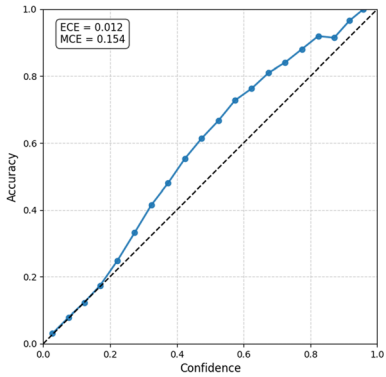

Fig. 10 Model calibration via prediction binning with 5% intervals, returning an ECE of 0.012 and a MCE of 0.154, with slight under-confidence in the mid-ranges, but overall returning a well-calibrated model

3.4 Pressure rankings for teams, formations, and position groups

By metricizing pressure such that the likelihood of a pass rusher creating pressure at any moment during a play can be quantified, the analysis can move beyond individual plays to more global insights. Teams can be ranked based on the average pressure their defense generates or the average pressure their offense allows. Similarly, pressure can be analyzed by formation or position group to understand how specific tactical choices perform against various opponents. These comparisons can be made across an entire season or limited to individual games.

While listing all possible pressure-based rankings is beyond the scope of this paper, illustrative examples are provided. Table 1 ranks all teams by the average pressure generated by their pass rush. Table 2 ranks teams by the average pressure allowed by their pass blocking units. Table 3 shows the average pressure generated by each position group of the Green Bay Packers. Table 4 ranks defensive formations of the Los Angeles Rams based on average pressure created.

3.5 Situation bias

At this point it has to be mentioned that, while the pressure metric can be considered beneficial in the areas of player and team performance, ranking individual players based on the pressure created is not straightforward. This is due to the fact that several factors can influence the efficiency of a pass rusher, some of which are position dependent. For example, when a Safety (the defensive players furthest away from the QB, typically acting as additional help to CBs in defending against WRs) is involved in the pass rush, the likelihood of the player achieving high pressure on the play is relatively big, the reason being that a “Safety Blitz” generally implies that the defense will have a surplus of pass rushers compared to the offensive line, allowing some of the pass rushers to get through to the QB without a blocker, which many times would be the Safety. The risk with this play being that a lack of players in the pass coverage part of the game could result in a completed pass with a significant gain in yardage by the offense if the QB picked up on the Blitz. Such a high-risk-high-reward type of play naturally favors the Safety in this pressure metric, regardless of how rarely the play is called. It is therefore of no use to compare such a player to a, for example,

Table 1 Average pressure created by the pass rush per team

Position	Team	Average pressure created (%)
1.	Cleveland Browns	13.5659
2.	Los Angeles Rams	13.5248
3.	Carolina Panthers	13.2895
4.	Baltimore Ravens	12.8947
5.	Las Vegas Raiders	12.8350
6.	Tennessee Titans	12.7755
7.	Miami Dolphins	12.7665
8.	Tampa Bay Buccaneers	12.7213
9.	Buffalo Bills	12.7045
10.	Arizona Cardinals	12.5961
11.	Dallas Cowboys	12.5301
12.	New England Patriots	12.4883
13.	Los Angeles Chargers	12.4344
14.	Jacksonville Jaguars	12.3899
15.	Pittsburgh Steelers	12.3825
16.	San Francisco 49ers	12.3543
17.	Denver Broncos	12.3377
18.	New York Jets	12.3297
19.	Chicago Bears	12.2973
20.	Kansas City Chiefs	12.2664
21.	Cincinnati Bengals	12.2644
22.	Seattle Seahawks	12.2551
23.	Washington Football Team	12.2527
24.	Minnesota Vikings	12.2362
25.	Philadelphia Eagles	12.2250
26.	Houston Texans	12.1708
27.	Green Bay Packers	12.1342
28.	Indianapolis Colts	12.1236
29.	New York Giants	12.1147
30.	Detroit Lions	11.8828
31.	New Orleans Saints	11.7707
32.	Atlanta Falcons	11.7210

Defensive End (a defensive lineman lining up towards one of the outside edges of the offensive line), who is generally faced with at least one pass blocker, and as a result has to fight to create pressure. This can be seen in the third ranking in the appendix, in which the average pressure created by each defensive position is being ranked. The SSL position (the Strong Safety on the left side of the field) has created an average amount of 41.7727%, while the LE and RE positions (Left End and Right End) have created 11.5220% and 11.0172% respectively. This difference in pressure created between the Left End and Right End, which are the same position group, only differing in the side of the field they are one, could be explained due to the QB's stance once the ball is in their hand. During a dropback, a QB faces the direction of their throwing arm, meaning a right-handed QB faces right, and a left-handed QB faces left. With this stance and the limited view a QB has due to the football helmet, the

Table 2 Average pressure allowed by the pass block per team

Position	Team	Average pressure allowed (%)
1.	Tampa Bay Buccaneers	11.0942
2.	Los Angeles Rams	11.3183
3.	Cleveland Browns	11.7475
4.	San Francisco 49ers	11.7479
5.	Pittsburgh Steelers	11.7703
6.	Dallas Cowboys	11.7806
7.	Cincinnati Bengals	11.9574
8.	New England Patriots	11.9781
9.	New York Giants	11.9829
10.	Tennessee Titans	12.0863
11.	Atlanta Falcons	12.0983
12.	Minnesota Vikings	12.1762
13.	Los Angeles Chargers	12.2579
14.	Green Bay Packers	12.2631
15.	Carolina Panthers	12.4082
16.	Washington Football Team	12.4492
17.	Buffalo Bills	12.5775
18.	Detroit Lions	12.6418
19.	Las Vegas Raiders	12.6870
20.	Baltimore Ravens	12.7007
21.	Arizona Cardinals	12.7521
22.	Kansas City Chiefs	12.7571
23.	New York Jets	12.8000
24.	Chicago Bears	12.8149
25.	Denver Broncos	12.8495
26.	New Orleans Saints	12.8792
27.	Jacksonville Jaguars	12.9339
28.	Miami Dolphins	12.9532
29.	Philadelphia Eagles	13.4621
30.	Seattle Seahawks	13.4665
31.	Indianapolis Colts	13.5094
32.	Houston Texans	13.7327

opposite side is known as the "blindside", since the QB is not able to see pressure from that direction. As a result, the offensive line typically attempts to protect this side of the field more, as a sack from the blindside could result in injuries or a fumble of the ball, since the QB is not able to prepare for the impact, unlike from the other side, from which the QB is able to see pass rushers coming more easily and is able to attempt to avoid them. With the large majority of the QBs in the NFL being right-handed, the left side of the offensive line is commonly filled with the better Offensive Linemen, which equates to the side that the Right Ends line up, resulting in less pressure being created, while the Left Ends are able to take advantage of the "weaker" throwing side of the QB's offensive line. A comparison within each also position appears to be difficult, as the amount of blockers that a player from a specific position faces does not remain constant. A player like Aaron Donald, who is considered

Table 3 Average pressure created by each position group for the Green Bay Packers

Position	Team	Average pressure created (%)
1.	SCBoR	30.9425
2.	LCB	22.9614
3.	SCBL	21.8651
4.	LLB	21.7633
5.	RILB	18.5515
6.	LOLB	15.9020
7.	LILB	15.4100
8.	RLB	14.9199
9.	NRT	14.5657
10.	ROLB	12.8220
11.	MLB	12.0788
12.	LE	11.2887
13.	LEO	10.9713
14.	RE	10.3622
15.	NLT	10.1703
16.	DLT	9.2391

Table 4 Average pressure created by each defensive formation for the Los Angeles Rams

Position	Team	Average pressure created (%)
1.	Cover-1	15.1962
2.	Red zone	14.3602
3.	Cover-0	14.1739
4.	Cover-3	13.7435
5.	Prevent	13.3908
6.	Cover-6	12.7865
7.	Quarters	12.6562
8.	2-Man	12.3701
9.	Cover-2	12.0175

to be one of the best pass rushers, would often times face two or even more blockers, as the opposition is typically aware of his strengths. His impact can be measured on a secondary level, as him drawing more pass blockers towards him, allows for his teammates to be faced with less or no pass blockers. The pressure metric should therefore only be considered in the analysis of individual plays of pass rushers, rather than using it as a comparison between players, as such results would be deemed insufficient or inaccurate.

3.6 Predicted motion pressure

To resolve the issue illustrated in Sect. 3.5, PMP was created. For this metric, two distinct steps had to be undertaken, requiring two different wKNN models, both using the same feature space, only differentiating in the outcome variables. Firstly, as mentioned in Sect. 2.3, the motion of a pass rusher was predicted using a wKNN model having the x-offset and y-offset of the pass rusher in the next frame compared to the current frame, along with the orientation of the pass rusher in the next frame as its three outcome variables. The train-test split was done identically to the split in

Sect. 3.1, with seven weeks having been used for training, and the remaining week for testing. Similarly to the wKNN used to predict the pressure of a pass rusher, each data point could be placed into the aforementioned 13-dimensional feature space. Again, a new frame could be placed into this feature space, the difference here being that each data point carried the x- and y-offsets and orientation for the next frame as its outcome variable. For this wKNN model, k = 50 was established to return the highest accuracy, meaning that a new frame would consider its 50 nearest neighbors and compute the average x- and y-offsets and orientation of these 50 nearest neighbors. This average value could then be considered the prediction for the motion of the pass rusher for this frame, the predicted x - and y-coordinates, as well as the new orientation of the pass rusher could then be computed straightforwardly, along with the speed, as mentioned in Sect. 2.3. A schematic visualization of the motion prediction can be seen in Fig. 11. Since the predicted values were averages computed from the values of the nearest neighbors, the motion of the pass rusher could be considered the average motion that would be made, based off of the motion of the 50 closest-to-identical pass rushers.

In the second step, the following frame could be analyzed in an identical manner to Sect. 3.1, with the position and speed of the pass rusher being replaced with the ones that were predicted in the previous step. All features were able to be computed in the exact same manner as for the pressure prediction. Therefore, this new data point could be placed in the same wKNN model, returning a percentage predicting the likelihood of pressure occurring on the play this predicted frame is a part of, labeled PMP. As a result, for any pass rusher for any frame, not only the pressure created by the pass rusher could be extracted, but also the pressure the pass rusher would have created, had they followed the motion that was predicted from the previous frame. This allowed for a straight-up comparison between the pressure that was actually created with the PMP, or, more precisely, a comparison between the actual pressure and the one that would have been created by the average player. As a result, the true pressure could be evaluated to be better or worse compared to the pressure that would have been created by the average pass rusher in this situation. A flowchart of how both the true pressure, as well as the predicted motion pressure are obtained from a starting frame can be seen in Fig. 12.

An illustration of the PMP in comparison to the actual pressure created for an entire play can be seen in Fig. 13. Here, the earlier mentioned pressure created by Indianapolis Colts' Deforest Buckner on play 4334 in the game against the Tennessee Titans in week 8 can be seen, along with his PMP for the play. Two decisive moments can be observed when comparing the true pressure with the PMP: firstly, the subtle spike in pressure between frames 11 and 13 (1.1–1.3 s into the play) argue for highly

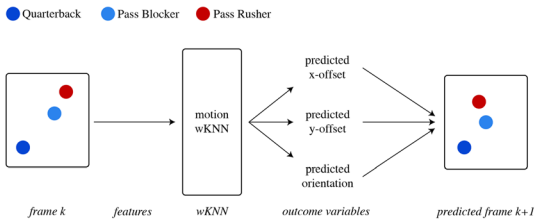

Fig. 11 Flowchart describing how the three KNN models predicting the motion of a pass rusher allow to go from a known frame k to predicting the following frame k+1

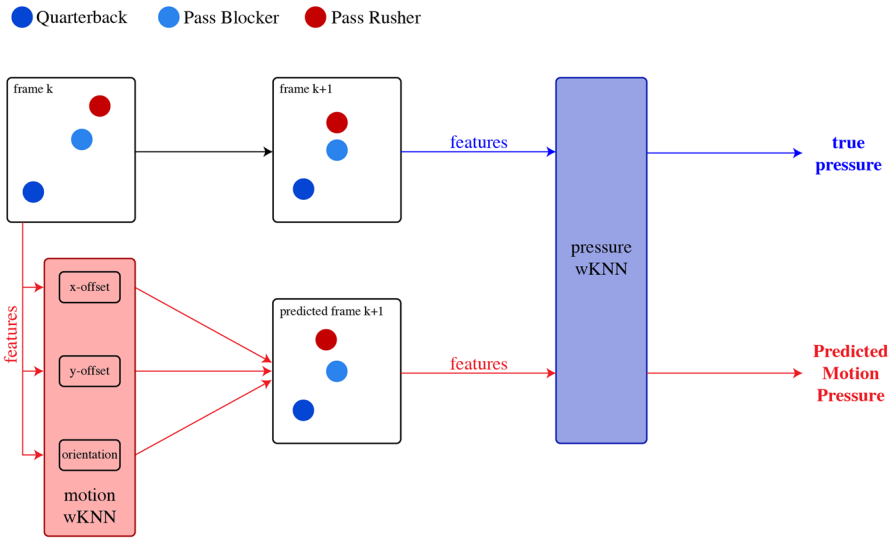

Fig. 12 Flowchart showing how to compute both the true pressure, as well as the PMP from a starting frame

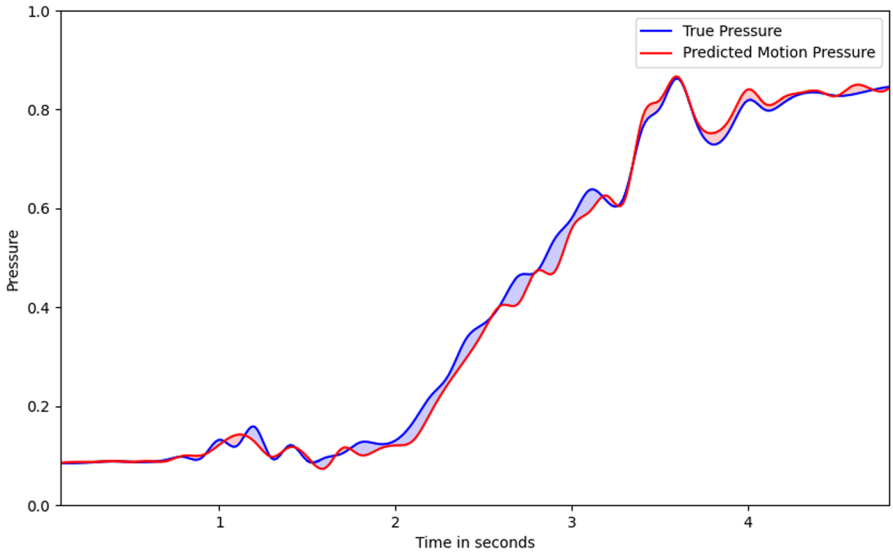

Fig. 13 Deforest Buckner's pressure and PMP on play 4334 in the game IND vs TEN in week 8

efficient move having been executed by Buckner, which can be explained by the true pressure being higher than the PMP, effectively setting Buckner up for the following successful pass rush and sack on Tannehill. Secondly, the continuous rise in pressure following the breakthrough of Buckner against the offensive line up is being accom-

panied by a rise in PMP, which is consistently lower than the actual pressure, apart from three frames in which both metrics are equal, indicating that Buckner was able to outperform the average pass rusher in this "free path" moment, showcasing his efficiency. The opposite can be detected during the peak of the pass rush, where the PMP is slightly higher than the true pressure, meaning that even during a successful pass rush, certain moments of improvement can be found. Overall, Buckner's delta between the true pressure and the PMP equaled 0.004722, meaning that Buckner not only did an obviously good job by creating pressure, but was also able to generate much more pressure than his predicted motion would have suggested for him to create. An visualization of key frames of this play along with the true pressure and PMP metrics can be seen in Fig. 16 in Appendix A.

3.7 PMP rankings

To demonstrate the PMP, two rankings have been created. Players can be ranked firstly based on their true pressure vs PMP delta on a single play, for which the top 20 highest can be seen in Table 5, and secondly, on the true pressure vs PMP delta average across all of the player's pass rushes, for which the top 20 highest can be seen in Table 6. Buckner's previously mentioned pass rush ranks 5828th out of 36,362 total recorded pass rushes.

Table 5 Top 20 highest PMP delta achieved on a play

Position	Team	Delta
1.	Michael Danna	0.04883603894312111
2.	Trey Hendrickson	0.041962586708640476
3.	Jerry Hughes	0.04191524832468291
4.	Juan Thornhill	0.04089150267975966
5.	Bobby Okereke	0.03859447767153873
6.	Anthony Barr	0.038262889198356964
7.	Eric Rowe	0.037280279263240845
8.	Randy Gregory	0.0365307956996862
9.	Shaquil Barrett	0.03618749729589374
10.	Anthony Chesley	0.03597757986785011
11.	Joe Tryon	0.035550063884667416
12.	Deion Jones	0.03543074465350535
13.	Fred Warner	0.035361593008124304
14.	Jonathan Allen	0.03423201878764173
15.	Jalen Reeves-Maybin	0.03417600687911676
16.	Malik Reed	0.03327918693088072
17.	Jihad Ward	0.033272903165749546
18.	Jeremiah Attaochu	0.033220963736407094
19.	Daron Payne	0.03307110829851111
20.	Elandon Roberts	0.03250975969670633

Table 6 Top 20 highest average PMP delta achieved

Position	Team	Delta
1.	Grant Stuard	0.02891725643670398
2.	Myles Hartsfield	0.025783979062341977
3.	Quandre Diggs	0.020268302281240685
4.	T.J. Edwards	0.01594719369247838
5.	Markus Bailey	0.014579252080835658
6.	Anthony Harris	0.013845159145955688
7.	Jordan Fuller	0.012549199001441858
8.	Jaylinn Hawkins	0.01221408368458371
9.	Chris Jackson	0.01127349229215182
10.	Davion Taylor	0.011179606028577455
11.	Tre Herndon	0.011006500480946571
12.	Donovan Wilson	0.010776934667900903
13.	Julian Blackmon	0.010597080593940614
14.	Jessi Bates	0.010188600069383977
15.	Dean Marlowe	0.009839486313577835
16.	Dorian O'Daniel	0.009470165881063947
17.	Drue Tranquill	0.009436809200549618
18.	Jalen Thompson	0.009323340639325994
19.	Anthony Chesley	0.00910579344279828
20.	Trayvon Mullen	0.0090608743787563

4 Discussion

Data science has established itself as a major part within professional sports, providing deeper insights to teams, coaches, and players into the finer margins of the game (Sarlis 2024). With the advancements in machine learning techniques, even more detailed discoveries in the field of sports analytics have become a possibility (Olaniyan et al. 2023). Pushing the limits of data science correlates directly with pushing the limits within any sport, as even the finest of margins of improvement in the performance of players and teams can make the difference between winning and losing. Improving the performance of players and teams is automatically followed by pushing the game itself into new heights.

Especially in the area of pass rush within American football, a more detailed metric measuring the pressure created not only provides a more intricate way for players and coaches to evaluate performances, but also provides insights into improvements that can be made to better protect the QB, directly correlating to the player's health and safety. The pressure metric elaborated on in Sect. 3.1 has created the possibility of analyzing every instance of play for every player involved in the pass rush, displaying exactly how much pressure was exerted on the QB. Coaches are able to retrieve specific moments in which the offense and/or defense acted well, therefore adjusting future game plans. Additionally, rankings can be created for position groups and formations, allowing for the evaluation of teams specifically.

This newly created metric is able to pick up minor pressure that the human eye may have missed out on, as the current metric of sacks/hits/hurry ups can only be applied to specific scenarios happening (Plessner and Bäumer 2006). The mental pressure that the QB faces cannot always be displayed by the current, classification-based metric, as minor pressure from many pass rushers, and the resulting presence of these players can be experienced by the QB and therefore influence their performance.

A new, pressure-based metric on its own can provide the aforementioned possibilities for in-depth analyses of teams, formations, or position groups, or for individual players on a play-by-play basis. As mentioned in Sect. 1.4.1 recent work by Nguyen et al., Hong et al., and Bachelder focus on novel ways to quantify pressure, all achieving excellent results in model accuracy. A direct comparison between the methodology described in this paper with the aforementioned ones is not deemed viable here, though, due to the fact that the nuances of each method can only be clearly detected when comparing individual plays. But while each model may be better suited for specific scenarios, there is no truly correct method to validate the result, since each model is built around the available, play-level, classification-based outcome of sacks, hits, and hurry-ups. However, the wKNN model described in this paper, due to its intuitive nature when comparing each moment to a set of "most-alike" moments and creating a weighted average, creates a straightforward and easily implementable metric analyzing pressure.

A common denominator of all pressure-related metrics, though, is the lack of ability to compare individual players against one another. To resolve this issue, PMP was created. On the one hand, PMP does allow for an additional analysis of players' individual performances on plays, as the true pressure created can be directly compared with the PMP for each frame, which, as mentioned earlier, theoretically could be used as a measurement for the pressure the average player would have created in that moment. It therefore can be established whether or not the movement that the pass rusher had done at any tenth of a second was better (if the true pressure is higher than the PMP) or worse (if the true pressure is lower than the PMP) compared to the average pass rusher. This comparison can be done of the entirety of the play, showcasing whether the pass rusher's decision making over the entirety of the play was net positive or net negative. At the same time, this again allows for more intricate observations in each player's performance, since the delta between true pressure and the PMP is not affected by how much pressure was created in that very moment. Regardless of whether the pass rusher faced two offensive linemen on a play or had a clear path to the QB, the difference in pressure signifies whether or not the movement of the pass rusher could be deemed above or below average. This transitions into the other advantage of the PMP metric, as not only can any player be evaluated on the quality of the movements at any given moment regardless of field position and pass rushing situation, but players can be compared among one another regardless of position. In the example of the Safety Blitz mentioned in Sect. 3.5, even though the Safety may have created a high amount of pressure, the PMP may show that at certain moments

the Safety could have made better decision and achieved even higher pressure, while in the case of Aaron Donald being double-teamed, more pressure, while still low, may have been created compared to the average player in his position, showcasing good movement and decision making.

The core concept of PMP with regard to predicting the motion of a player falls within the research topic of "ghosting", in which several advancements have been made in recent year, as described in Sect. 1.4.2. However, many of these methodologies are based around intricate, computationally heavy deep learning models, while PMP allows for a straightforward and intuitive way of predicting the motion of the player. Similarly to the wKNN model that predicts the likelihood of pressure, PMP computes the motion of the player at any moment from the weighted average of a set of "most-alike" moments, which can directly be interpreted as the motion that the average pass rusher would have undertaken in this situation. As a result, this allows for the direct comparison in true pressure and the hypothetical pressure from the predicted motion.

While the time frame of one tenth of a second appears to be low in regard to predicting the motion of players into the future, the PMP's main goal has been to showcase the decision making of pass rushers and its effects on the play. With plays generally being relatively short (only lasting a few seconds), any split-second decision can make the difference between a sack and being stuck in front of the offensive line. The PMP metric allows for the discovery of players' performances that may not have been straightforward to see due to outside factors keeping the pass rusher's pressure low, yet undermining the player's strong decision-making and anticipation skills. To many players, this can have large implications on their overall evaluation by the team, discovering new strengths of certain players. With this in mind, it should be noted that decisions regarding any player and their role on a team should not be made purely based on data science metrics. These, and any other metrics should rather be used as an aid to the coaches, staff, and the players themselves.

The pressure metric and PMP have been showcased to provide more intricate details into the game of American football, more specifically the pass rush. Teams, coaches, and players may use these new metrics for a better evaluation of their performances, giving them insights into the areas in which they may seek improvements, while coaches may use these metrics to gain an advantage in strategizing by adapting certain game plans that favor their team in regards to pressure created/allowed. From this point onward, it is the analysts of each individual team that can use these two metrics for individual team and player performance evaluation.

Appendix A additional figures

See Figs. 14, 15 and 16.

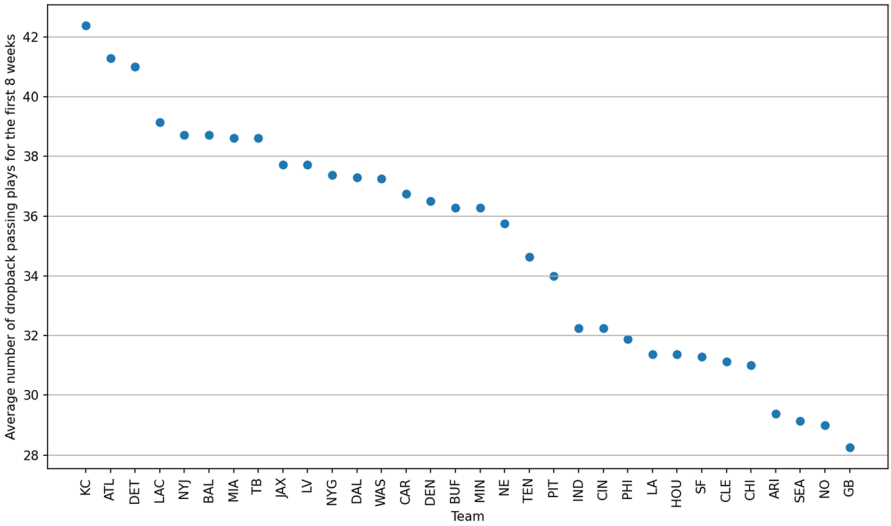

Fig. 14 Average number of passing plays for the first 8 weeks per team

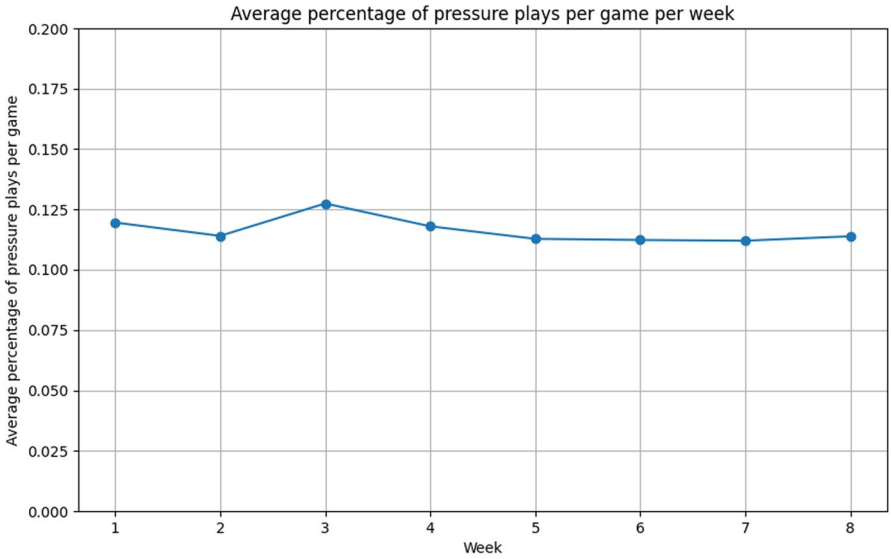

Fig. 15 Average percentage of pressure exerted by the defense per game for all games of a week across all eight weeks

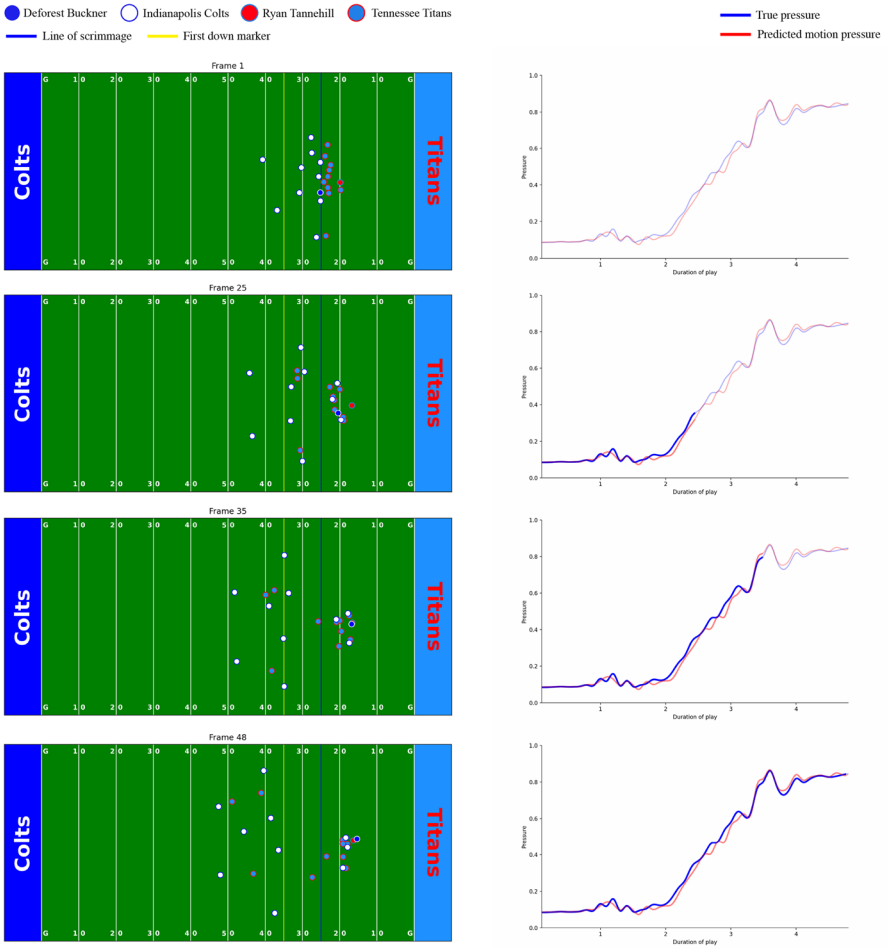

Fig. 16 Visualization of key frames of play 4334 of the game IND vs TEN with focus on Deforest Buckner's created pressure and PMP

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Author contributions C.P. came up with the original methodology and did the computation/coding. The main manuscript was written by C.P. and S.A., D.V. did important reworking with regard to the methodology and the discussion. C.P. prepared figures 1, 5, 6, 7, 9, 10, 12, 13, A1, A2 and A3, S.A. prepared figures 3 and 11, D.V. 2, 4, and 8. All authors reviewed the manuscript.

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Data availability The raw data is publicly available. The generated data with regard to feature selection can be obtained on request from the corresponding author.

Material availability Not applicable.

Code availability The code can be obtained on request from the corresponding author.

Declarations

Conflict of interest The authors state no Conflict of interest.

Ethics approval and consent to participate Not applicable.

Consent for publication Consent has been given by all authors.

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